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Source: *Oxford Economic Papers*, Vol. 43, No. 4 (Oct., 1991), pp. 667-683

Published by: Oxford University Press

Stable URL: <https://www.jstor.org/stable/2663415>

Accessed: 11-05-2026 14:40 UTC

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THE MEASUREMENT AND DECOMPOSITION OF COST-INEFFICIENCY: THE TRANSLOG COST SYSTEM*

By SUBAL C. KUMBHAKAR

1. Introduction

IN recent years there has been a growing interest in the estimation of efficiency in production. Although the basic idea of measuring efficiency goes back to Farrell (1957), econometric estimation of it is quite recent (see Aigner et al. (1977), Forsund et al. (1980), Schmidt (1985–6), and the most recent survey by Bauer (1990)). In the frontier methodology, efficiency of an individual production unit is measured by comparing its performance to a standard—either the cost, profit or the production frontier. Forsund et al. (1980) defined cost frontier as the minimum cost function for given level of output and input prices. Similarly, the profit frontier is the maximum profit function given input and output prices. Finally, the production frontier is the locus of maximum possible output given the level of inputs. The notion of these frontiers is consistent with the optimizing behavior of producers where concepts of maximality (in production and profit functions) and minimality (in cost function) are emphasized.

In econometric estimation of production, cost, and profit frontiers one has to choose a functional form. The advantage of using a restrictive technology such as Cobb-Douglas lies in: (i) the specification and estimation of different types of inefficiencies, and (ii) decomposition of cost- (profit-) inefficiency into different components of inefficiencies. The disadvantage is the loss of flexibility. Since the technology is rarely known, it is desirable to consider a flexible production, cost, or profit function to represent the underlying technology so as to test some restrictive specifications. When it comes to specifying a flexible functional form for the unknown technology the problem is whether one can deal with the above issues satisfactorily.

If one starts with the translog cost function—the most widely used flexible functional form in a cost minimizing framework—the common approach is to use the cost function and the associated share equations (deleting one) and use Zellner's seemingly unrelated regression technique as the basis of estimation (see Christensen and Greene (1976)). This, however, may not be desirable especially when there are technical and allocative inefficiencies. Neglecting technical inefficiency in a cost function—where output is exogenous—may not affect consistency property of the parameter estimates, except for the intercept. This is because Farrell's definition of technical inefficiency is independent of input prices. However, exclusion of allocative inefficiency may result in inconsistent parameter estimates since allocative inefficiency is price dependent.

* I thank C. L. Gilbert, M. B. Gregory, Knox Lovell, Peter Schmidt, and two anonymous referees for their comments on an earlier version of the paper.

On the other hand, if consistent estimates of the parameters of the cost function are somehow obtained, one can use, in principle, the Kopp-Diewert (1982) and the Zieschang (1983) algorithm to decompose cost of inefficiency into costs of allocative and technical inefficiencies. But whether one obtains consistent estimates or not depends crucially on the specification of technical and allocative inefficiencies in the cost function. This is why specification of these inefficiencies in the cost function is so important.¹

This paper has two objectives. First, we consider the translog cost function which incorporates both technical and allocative inefficiencies. Second, a theoretically consistent linkage between allocative inefficiency—as measured by the errors of the cost share equations—and cost therefrom is provided. To achieve these goals we estimate the translog cost system with inefficiencies first, and then disentangle cost-inefficiency into measures of technical and allocative inefficiencies from the composite error term in the cost function. Based on some distributional assumptions on the error components, we develop the maximum likelihood (ML) method to obtain consistent parameter estimates. These estimates are then used, in the second step, to estimate costs of technical and allocative inefficiencies for each observation. Thus the paper deals with the problems of estimation as well as decomposition of cost-inefficiency. As an illustration of the methodology, we use data on ten US airlines observed over 1970–8.

2. Specification of the structure of cost-inefficiency

Since the cost function of a firm summarizes all the economically relevant aspect of its technology, econometricians often specify models based on cost function instead of production functions. The choice of one over the other depends on the behavioral assumption on the part of the producers and/or availability of data. The cost function approach is widely used to model the technology of a firm operating in regulated environments such as electricity, telephone, airlines, etc. Outputs of firms in these industries are assumed to be exogenous. Since input prices are also exogenous under competitive factor market assumption, one can obtain consistent parameter estimates by using only the cost function in estimation. However, one problem with this formulation is that each producer is assumed to be efficient. The only error in this model is white noise and possible model mis-specification.

The traditional cost function is modified to the extent that a cost-minimizing firm may not be able to produce a given level of output (with given input prices) at a minimum possible cost (the cost frontier) if technical and allocative inefficiencies are present. The former can arise due to poor management, low quality of inputs, etc., while the latter is due to optimization error, i.e., failure to use inputs by equating the ratio of input prices to their marginal products.

¹ The problem is somewhat simpler if one deals with the production functions for which the dual cost functions can be algebraically derived. In such cases costs of technical and allocative inefficiencies can be explicitly derived as functions of technical and allocative errors.

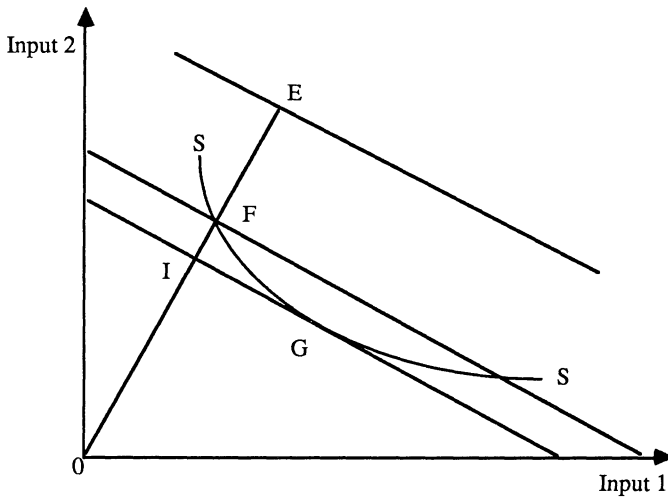


FIG. 1.

Thus, the sources of errors in the production and cost are: (a) technical inefficiency, resulting from the failure to produce maximum possible output, given a set of inputs, (b) allocative inefficiency, arising out of the choice of sub-optimal input-proportions, given input prices, and (c) white noise—representing exogenous shocks, luck factors, etc., which are not in the control of any firm. The last component may also include approximation error if the underlying true cost function is not translog. Consequently, the errors in the cost function must reflect these three types of errors. These lead us to the following specification of the cost function

$$\ln C = \ln C^* + \varepsilon \quad (1)$$

where $\ln C$ = log of total cost, $\ln C^*$ = log of minimum cost (which defines the deterministic cost frontier²), and $\varepsilon = \ln C_\tau + \ln C_a + \ln C_v$. $\ln C_\tau$ and $\ln C_a$ represent increase in log cost due to technical and allocative inefficiencies respectively. $\ln C_v$ is white noise which may increase or decrease cost. Both $\ln C_a$ and $\ln C_\tau$ are non-negative since technical and allocative inefficiencies increase cost.³

The appearance of C_τ and C_a in log linear form in (1) is explained with the help of Figure 1. Let the observed input combination be denoted by the point E . The isoquant SS indicates the level of output obtained from the using input combination in E . Comparison of E with the technically efficient point F shows excess use of both the inputs by the same proportion.⁴ Moreover, the excess

² The cost frontier can be made stochastic by adding the white noise component $\ln C_v$ to it.

³ It is to be noted that regulatory constraints, if not properly modelled, may affect cost. This effect, if any, is included in ε .

⁴ This is often called input saving measure of technical inefficiency. See Forsund and Hjalmarsson (1987) for more on this point.

use of inputs due to technical inefficiency is independent of input prices. Consequently cost of technical inefficiency, C_τ —measured by comparing costs at points E and F —is proportional to the minimum or frontier cost, C^* , given by the budget line passing through the point G which is both technically and allocatively efficient. Similarly, cost of allocative inefficiency can be measured by comparing costs at points F and G . Since costs at points I and G are the same, cost of allocative inefficiency, C_a , is also proportional to the minimum cost, C^* . This can also be seen from the fact that the points E , F , and I are on the same ray through the origin. Thus actual cost (disregarding white noise component) can be expressed as a product of C^* , C_τ and C_a . As a result of this both $\ln C_\tau$ and $\ln C_a$ appear in additive form in (1). Finally, the white noise component is added to (log) actual cost in deriving the expression in (1).

The cost function in (1) can be directly estimated after specifying a functional form on $\ln C^*$ and making some distributional assumptions on $\ln C_v$, $\ln C_a$, and $\ln C_\tau$. This gives consistent parameter estimates if allocative inefficiency is independent of input prices. However, as we have noted, that is not true. Furthermore, $\ln C_a$ and $\ln C_\tau$ can not be separately identified from (1). But one can separate out costs of technical and allocative inefficiencies if the model is expanded to include the cost share equations; and some additional assumptions are made linking $\ln C_a$ with allocative inefficiency. This requires a formal definition of technical and allocative inefficiencies.

Forsund et al. (1980) defined a firm to be allocatively inefficient if $F_i/F_j \neq w_i/w_j$ for all $i, j (i \neq j)$, where F_i is the marginal product of input X_i and w_i is the price of X_i . This, by using the duality results on production and cost (Shephard's lemma), is equivalent to $C_i/C_j \neq X_i/X_j$ for $i \neq j$ where C_i is the partial derivative of C with respect to input price w_i . The latter definition implies that the actual shares are different from the optimum shares, i.e.,

$$S_i = S_i^* + U_i, \quad i = 1, \dots, n \quad (2)$$

where S_i is the actual share and S_i^* is the optimal share of input i in the total cost. Since the cost shares must sum to unity, $\sum U_i = 0$. A non-zero value of U_i for an input i is an indication of allocative inefficiency. Thus $\ln C_a$ and U_i are obviously related.⁵ The problem is how to relate U_i to $\ln C_a$ since no closed form solution is possible for the translog cost function.⁶ Since both positive and negative U_i increase cost, Schmidt (1984) proposed a specification

$$\ln C_a = U'AU \quad (3)$$

where A is a *pre-specified* positive semi-definite matrix and $U = (U_1, U_2, \dots, U_n)'$. In this specification the cost of allocative inefficiency is a

⁵ Greene (1980) considered a translog system that allowed technical and allocative inefficiencies. He acknowledged that U_i and $\ln C_a$ are related but assumed them to be independent.

⁶ For production functions with self dual cost functions (e.g., Cobb-Douglas, CES, etc.) $\ln C_a$ can be explicitly derived from the Forsund et al. (1980) definition of allocative inefficiency. The share equations, if desired, can also be obtained and these are well defined functions of allocative inefficiency and the parameters of the cost function (see Schmidt and Lovell (1979) and Kumbhakar (1987) for the Cobb-Douglas case).

weighted average of squared allocative mistakes represented by the errors in the share equations. Its interpretation is quite simple. First, cost is increased as the magnitude of allocative errors increases. Second, cost is increased by the same proportion for making allocative mistakes in either direction. However, the choice of $\ln C_a$ in (3) is arbitrary except for the requirement that $\ln C_a \geq 0$ for $U_i \neq 0$ for at least one i .

The choice of A should be made in such a way that in addition to the above requirement, one can get the share equations by applying Shephard's lemma to (1) or should obtain the cost function, except for C_v , by integrating the share equations in (2). This requirement is missing in Schmidt (1984) (see also Bauer (1985), Ferrier and Lovell (1990)). The arbitrariness of A can be eliminated by imposing the above integrability condition. This is considered in terms of a translog cost function where $\ln C^*$ is

$$\ln C^* = \alpha_0 + \delta_y \ln y + \alpha' \ln w + \frac{1}{2} \ln w' B \ln w + \frac{1}{2} \delta_{yy} \ln y^2 + \lambda' \ln w \ln y + \frac{1}{2} \ln q' \Delta \ln q + \ln w' D \ln q + \psi' \ln q \ln y + \theta' \ln q \quad (4)$$

where y is output, w is the vector of n input prices, and q is a vector of m exogenous variables. In addition to the symmetry restrictions on the matrices B and Δ , we impose the following restrictions: $\alpha' \iota = 1$, $B' \iota = 0$, $D' \iota = 0$, $\lambda' \iota = 0$ where ι is a vector of ones. These restrictions make the cost function in (4) linear homogeneous in input prices.

By applying Shephard's lemma to (1) we get

$$S_i = S_i^* + \partial \ln C_a / \partial \ln w_i \quad (5)$$

where

$$S_i^* = \alpha_i + \sum_{j=1}^n \beta_{ij} \ln w_j + \sum_{k=1}^m d_{ik} \ln q_k + \lambda_i \ln y, \quad i = 1, \dots, n. \quad (6)$$

$\partial \ln C_a / \partial \ln w_i$ has to be equal to U_i since the share equations in (2) are to be identical to those in (5). This will impose some additional constraints on A . Using the constraint $\Sigma U_i = 0$, the relation in (3) can be rewritten as $u'Ku$ where $u = (U_1, \dots, U_{n-1})'$ and K is an $(n-1) \times (n-1)$ matrix. Its elements are well defined functions of the elements of A . It can be shown (see Appendix 1) that $\partial(u'Ku)/\partial \ln w_i = U_i$ for $i = 1, \dots, n-1$ if

$$\sum_{j=1}^{n-1} k_{ij} \beta_{ij} = -\frac{1}{2},$$

$$\sum_{j=1}^{n-1} k_{ij} \beta_{ij} = 0 \text{ for all } i \neq l, \quad i = 1, \dots, n-1 \quad (7)$$

where k_{ij} are the elements of K and $k_{ij} = k_{ji}$. In matrix form, these conditions can be written as $KB_0 = -\frac{1}{2}I$ where B_0 is the B matrix with the last row and column deleted and I is an identity matrix of order $(n-1)$. Thus the solution

for K is

$$K = -\frac{1}{2} B_0^{-1} \quad (7a)$$

and there is no additional parameter to be estimated in K .⁷ The only constraint is that K has to be a positive definite matrix and, therefore, B_0^{-1} negative definite. Using (7) and (7a) the final form of $\ln C_a$ is given by⁸

$$\ln C_a = -\frac{1}{2} u' B_0^{-1} u. \quad (8)$$

Another question often raised in this context is whether the share equations contain technical error. The answer to this question depends on whether output is exogenous or not. In a cost-minimizing setup output y is treated as exogenous and the inputs endogenous. Thus y is no longer a random variable and can not possibly be affected by any errors.⁹ The empirical analysis of the present study is based on a sample of US airlines. During the period of our observation (1970–8) the airline industry was regulated. Following the literature on regulated industries we assume that the objective of the airlines is to minimize cost given output. Exogeneity of output follows from the fact that the industry is demand constrained and produced output cannot be stored. Furthermore, there are different types of regulatory constraints which make unconstrained profit maximization impossible so far as the choice of factors affecting output is concerned.

3. The likelihood function and method of estimation

Estimation of the model in (4), (5), and (8) requires specification of distributional assumptions on $\ln C_v$, $\ln C_\tau$, and u . Here we drop the last share equation. Regarding distribution of the error vector we assume: (i) u is distributed independently and identically over firms as multivariate normal with zero mean and constant covariance Σ , (ii) $\ln C_v$ is i.i.d. $N(0, \sigma_v^2)$, (iii) $\ln C_\tau$ is i.i.d. $N(0, \sigma_\tau^2)$ variable truncated at zero from below, and (iv) $\ln C_v$, $\ln C_\tau$ and u are assumed to be mutually independent.

With these distributional assumptions the joint probability density function (pdf) of (ε, u) , $f(\varepsilon, u)$ is

$$f(\varepsilon, u) = f(u)f(\varepsilon|u)$$

⁷ This result makes the present model different from the one used by Bauer (1985), and Ferrier and Lovell (1990). They assumed K to be diagonal and estimated the elements of K as separate parameters. Moreover, the integrability condition is not satisfied in their models.

⁸ Note that the form of $\ln C_a$ in (8) is based on the specification of it in (3) which is only an approximation of the 'true' relation. If the true relationship is known as in the case of self dual production functions, the exact relationship between the observed and optimum cost shares can be obtained from (5).

⁹ On the other hand, if y is endogenous, it will be affected by technical inefficiency. But in this case, estimation of the cost function is not appropriate even if the underlying production function is homothetic (i.e., when the elements of λ and ψ are all zero). This is because y and ε will be correlated (via technical and allocative errors).

where $f(u)$ is the marginal pdf of u and $f(\varepsilon|u)$ is the conditional pdf of ε given u . The pdf of u , $f(u)$ is

$$f(u) = \frac{1}{(2\pi)^{(n-1/2)}|\Sigma|^{1/2}} \exp\left\{-\frac{1}{2}u'\Sigma^{-1}u\right\}. \quad (9)$$

Now $f(\varepsilon|u) = f(\ln C_\tau + \ln C_v)$ since $\ln C_a$ becomes known given u . It can be shown that

$$f(\ln C_\tau + \ln C_v) = \frac{2}{(\sigma_v^2 + \sigma_\tau^2)^{1/2}} \frac{1}{(2\pi)^{1/2}\Phi(\mu/\sigma)} \exp\left\{-\frac{z^2}{2(\sigma_v^2 + \sigma_\tau^2)}\right\}$$

where $\sigma = \sigma_v\sigma_\tau/(\sigma_v^2 + \sigma_\tau^2)^{1/2}$, $\mu = z\sigma_\tau^2/(\sigma_v^2 + \sigma_\tau^2)$, $\Phi(\cdot)$ is the cumulative pdf of a standard normal variable and $z = \varepsilon - \ln C_a$. Thus the log-likelihood function for a single firm can be written as

$$L = \ln f(u) + \ln f(\varepsilon|u) = \text{const.} - \frac{1}{2}\ln|\Sigma| - \frac{1}{2}(u'\Sigma^{-1}u) + \ln \sigma - \ln \sigma_v \\ - \ln \sigma_\tau - \ln \Phi\left(\frac{\mu}{\sigma}\right) - \frac{z^2}{2(\sigma_v^2 + \sigma_\tau^2)} \quad (10)$$

where ε , u , and $\ln C_a$ are to be replaced by their observable counterparts from (1), (2), (8), and (4). The log-likelihood function for a sample of F firms observed over T years can be obtained from (10) by summing it up over the firms for all the years. By doing so we are implicitly assuming that $\ln C_v$, $\ln C_\tau$ and u are i.i.d. over firms for all the years¹⁰ in addition to the assumptions (i)–(iv) in the beginning of this section. The likelihood function thus obtained can be maximized to get the ML estimates of α_0 , the elements of α , θ , B , λ , Δ , D , ψ , Σ , and δ_y , δ_{yy} , σ_v^2 , σ_τ^2 . However, the computational burden can somewhat be reduced by concentrating the likelihood function with respect to Σ . Since the ML estimate of σ_{ij} , the (i, j) th element of Σ , is

$$\sigma_{ij} = \frac{1}{FT} \sum_f \sum_t u_{if_t} u_{jf_t} = \frac{1}{FT} \sum_f \sum_t (S_{if_t} - S_{if_t}^*)(S_{jf_t} - S_{jf_t}^*) \quad (11)$$

where subscripts f and t index firm and time ($f = 1, \dots, F$, $t = 1, \dots, T$), the ML estimates of α_0 , elements of α , θ , B , λ , Δ , D , and ψ , δ_y , δ_{yy} , σ_v^2 , and σ_τ^2 can be obtained by maximizing the concentrated log-likelihood function

$$L_c = \text{const.} - \frac{1}{2} FT \ln|\Sigma| + FT \ln \sigma - FT \ln \sigma_v - FT \ln \sigma_\tau - \sum_f \sum_t \ln \Phi\left(\frac{\mu_{ft}}{\sigma}\right) \\ - \sum_f \sum_t \frac{z_{ft}^2}{2(\sigma_v^2 + \sigma_\tau^2)} \quad (12)$$

where σ_{ij} in Σ are replaced by relation (11). The ML estimates of σ_{ij} are obtained from (11) where $S_{if_t}^*$ is replaced by its estimate from (6). These ML estimates

¹⁰ In Schmidt and Sickles (1984) and Kumbhakar (1987, 1988) technical inefficiency is assumed to be time invariant. This assumption is relaxed in Cornwell et al. (1990) and Kumbhakar (1990).

can be used to find the cost of allocative inefficiency from (8) where the elements of u are to be replaced by the residuals of (6). Finally, $\ln C_\tau$ can be estimated from the mean of $\ln C_\tau$ conditional on ε , viz.,

$$\ln C_\tau = \mu + \sigma \frac{\phi\left(\frac{\mu}{\sigma}\right)}{\Phi\left(\frac{\mu}{\sigma}\right)} \quad (13)$$

where $\phi(\cdot)$ is the pdf of a standard normal variable.

4. Empirical results

In this study we use panel data on ten US airlines observed from 1970 through 1978—during the period when the airline industry was regulated. The data set is developed by the Christensen Associates. Details about construction and sources of these data can be found in Tretheway and Windle (1983). The data set has also been discussed and analysed in Caves et al. (1983, 1984). The source of the data is the Civil Aeronautic Board's Form 41 filed annually by each airline.¹¹ The airlines included in this study are American, Delta, Eastern, Frontier, Ozark, Piedmont, TWA, United, US Air and Western.

The data set contains information on quantities and costs of five categories of inputs. These are labor, fuel, flight equipments, ground property, and materials. We decided to reduce the number of inputs to three so as to keep the number of parameters reasonably low. Thus we aggregate flight equipments, ground property, and materials to form an index of capital using multilateral index procedure (see Caves et al. (1981, 1982)). Labor is an index of 15 categories of employees. Fuel index is based on total gallons of fuel consumed. The prices of these inputs are obtained by dividing the costs of each by the corresponding quantity. Output is an index of four categories of outputs; viz., revenue passenger-miles (RPM) of scheduled service, RPM of charter service, revenue ton-miles (RTM) of mail, and RTM of all other freight, obtained by using the multilateral index procedure. Total cost is the sum of costs of the above mentioned five categories of inputs. We also include two control variables, viz. average stage length and average load factor, one network variable (points served), and time as a shift variable in the model. Average stage length is the average distance between takeoffs and landings and average load factor is the ratio of seat miles sold to seat miles flown. Time is measured in years and points served is the actual number of airports served. The variables in q are average stage length, load factor, number of points served, and time. All the variables in the cost function are normalized to unity for Delta airline in 1977.

$$\varepsilon_{ft} = \ln C_{\tau f} + \ln C_{a ft} + \ln C_{v ft} + \gamma_f H_f \quad (14)$$

¹¹ I thank Robert Windle of Christensen Associates for providing these data.

where H_f are airline-specific dummy variables and γ_f are their coefficients. In $C_{\tau ft}$, $\ln C_{aft}$, and $\ln C_{vft}$ are assumed to be independent over airlines and time as well as independent among themselves. However, panel nature of the data set is exploited by introducing the airline-specific effects which are known as fixed-effects model in the panel data literature. The estimation is carried out by dropping the share equation for capital. The resulting estimates are reported in Appendix 2. Since the total cost and the regressors (except for time) are in natural logarithms, the first order coefficients can be interpreted as cost elasticities for Delta in 1977 (the point of normalization). Thus cost elasticities of labor and fuel are .376 and .177, respectively in Model I. These elasticities can also be interpreted as the cost share of labor and fuel. Time, stage length, load factor, and points served are all negatively related to total cost for Delta in 1977. Due to exogenous technical progress total cost decreased over time at the rate of 1.70%. A 1% increase in load factor, for example, implies a reduction in total cost by 11%. Other coefficients can be similarly interpreted in Models I–IV. The estimates of σ_τ are found to be statistically significant in all the models. These results provide direct test for the presence of technical inefficiency since the asymptotic t tests fail to reject the hypothesis $\sigma_\tau = 0$ in Models I–IV. However, the magnitude of σ_τ is smaller relative to σ_v in all the models.

In addition to the general model outlined in section 2, which we call Model I, some special cases of Model I are also considered. Model II corresponds to the homothetic translog case, and Model III assumes constant returns to scale. Finally, Model IV considers the translog model with Hicks neutral technical progress. Thus Model II is obtained from Model I by imposing the following restrictions

$$\lambda_i = 0, \text{ and } \psi_j = 0. \quad (15)$$

Similarly, the restrictions on Model I to get Models III and IV are

$$\lambda_i = 0, \psi_j = 0, \delta_{yy} = 0, \text{ and } \delta_y = 1 \quad (16)$$

$$\delta_{jt} = 0, \psi_{ty} = 0, d_{it} = 0 \quad (17)$$

respectively. δ and d in (17) are the elements of Δ and D matrices in (4).

Restrictions on the parameters implied by Models II–IV are statistically tested using the likelihood ratio (LR) test. The LR test statistics for these models are reported in Table 1. It can be seen from Table 1 that the hypothesis of homotheticity (Model II) is rejected at 1% level of significance. Similarly, the constant returns to scale (Model III), and Hicks' neutral technical progress (Model IV) hypotheses are also rejected at 1% level of significance. These results show that the present data support the unrestricted translog model—Model I.¹²

Estimates of cost of technical inefficiency for each of these airlines under Model I are reported in Table 2. These estimates for each airline over all the

¹² Theoretical properties of the cost function, viz., symmetry and homogeneity of degree one in input prices, are not empirically tested. Thus the results are conditional on symmetric and homogeneous translog cost structure.

TABLE 1
Test Statistics

<i>Model</i>	<i>Number of Restrictions</i>	<i>Chi-squared Test Statistic</i>	<i>Critical Value at 1%</i>
II	6	58.62	16.81
III	8	87.28	20.09
IV	7	29.42	18.47

TABLE 2
Estimates of Costs of Technical Inefficiency (Model I)

<i>Airline</i>	<i>1970</i>	<i>1971</i>	<i>1972</i>	<i>1973</i>	<i>1974</i>	<i>1975</i>	<i>1976</i>	<i>1977</i>	<i>1978</i>	<i>Mean</i>
American	5.09	5.07	5.10	5.16	5.19	5.22	5.22	5.24	5.21	5.17
Delta	4.59	4.66	4.66	4.76	4.76	4.82	4.81	4.82	4.78	4.74
Eastern	4.84	4.88	4.88	4.99	5.03	5.02	5.06	5.04	5.00	4.97
Frontier	4.10	4.12	4.11	4.15	4.21	4.24	4.25	4.24	4.33	4.19
Ozark	4.13	4.12	4.15	—*	4.29	4.39	4.29	4.30	4.29	4.24
Piedmont	4.10	4.12	4.16	4.21	4.23	4.31	4.27	4.25	4.28	4.21
TWA	5.13	5.05	5.11	—*	5.24	5.28	5.31	5.35	5.29	5.22
United	5.00	5.03	5.02	5.03	5.11	5.19	5.20	5.22	5.12	5.10
US Air	4.30	4.40	4.43	4.54	4.54	4.59	4.60	4.60	4.63	4.51
Western	4.38	4.42	4.43	4.47	4.53	4.55	4.53	4.55	4.55	4.49

* These observations are deleted from the sample due to strikes in excess of 25 days.

years are obtained from equation (13). Instead of reporting $\ln C_t$, we calculated the percentage increase in costs due to technical inefficiency from the expression $\{\exp(\ln C_t) - 1\}100$. Thus the entries in Table 2 for each airline can be interpreted as the percentage increase in cost due to technical inefficiency. Comparing across airlines, we do not find much difference in cost of technical inefficiency. Frontier and Piedmont airlines are found to be technically most efficient in 1970. Their costs were 4.10% higher due to technical inefficiency. The least efficient airline in the sample in 1970 was TWA whose cost was 5.13% higher for being technically inefficient. When compared over time, the results of Table 2 indicate a tendency of technical inefficiency to increase for most of the carriers, especially during 1970–5. However, this tendency was reversed after 1977 except for Eastern, Frontier, Piedmont, and US Air. In 1978 cost of technical inefficiency range from 4.28% (Piedmont) to 5.29% (TWA).

Since none of the airlines in the sample was technically efficient during the period 1970–8 it may be interesting to compare performance of the other airlines relative to Frontier and Piedmont which were technically most efficient in 1970. This comparison shows, given everything else, the cost of American relative to Frontier to be .99% higher in 1970. Similarly, Eastern, relative to Frontier, was above the cost frontier by .74% due to technical inefficiency alone.

Rankings of the airlines in terms of their relative performances in different years are found to be quite similar.

Table 3 reports the estimates of percentage increase in costs due to technical inefficiency under the restrictive Models II-IV (which are rejected in favor of Model I). Here we report only the mean (over time), maximum and minimum values for each airline. These estimates corresponding to Models II-IV are somewhat higher compared to Model I.

Percentage increase in costs due to allocative inefficiency for each airline at every year is calculated using the relation $\{(\exp(\ln C_a) - 1)100\}$. In all the models, we find allocative inefficiency to be so small that estimates of costs of

TABLE 3
Estimates of Costs of Technical Inefficiency

<i>Airline</i>		<i>Model II</i>	<i>Model III</i>	<i>Model IV</i>
American	Mean	5.80	6.13	7.10
	Max.	5.92	6.38	7.23
	Min.	5.63	5.75	6.89
Delta	Mean	5.25	5.11	6.10
	Max.	5.37	5.28	6.28
	Min.	5.02	4.73	5.80
Eastern	Mean	5.57	5.78	6.61
	Max.	5.69	6.01	6.78
	Min.	5.37	5.41	6.36
Frontier	Mean	4.61	4.72	5.16
	Max.	4.78	4.94	5.41
	Min.	4.49	4.53	4.98
Ozark	Mean	4.57	4.61	5.33
	Max.	4.75	4.97	5.69
	Min.	4.40	4.37	5.07
Piedmont	Mean	4.53	4.54	5.28
	Max.	4.67	4.69	5.50
	Min.	4.36	4.04	5.05
TWA	Mean	5.96	6.47	7.20
	Max.	6.13	6.80	7.46
	Min.	5.75	6.01	6.80
United	Mean	5.62	5.86	6.91
	Max.	5.79	6.19	7.14
	Min.	5.42	5.41	6.72
US Air	Mean	5.06	4.93	5.78
	Max.	5.23	5.28	6.03
	Min.	4.75	4.51	5.36
Western	Mean	5.08	5.39	5.51
	Max.	5.18	5.57	5.63
	Min.	4.94	5.16	5.29

TABLE 4
*Estimates of Mean Costs of Allocative Inefficiency**

<i>Airline</i>	<i>Model I</i>	<i>Model II</i>	<i>Model III</i>	<i>Model IV</i>
American	.090 (.051)	.084 (.049)	.080 (.049)	.083 (.045)
Delta	.092 (.099)	.242 (.191)	.232 (.188)	.117 (.144)
Eastern	.056 (.059)	.056 (.049)	.060 (.056)	.101 (.072)
Frontier	.048 (.035)	.112 (.067)	.113 (.067)	.078 (.058)
Ozark	.072 (.079)	.069 (.083)	.072 (.084)	.260 (.271)
Piedmont	.288 (.439)	.290 (.483)	.294 (.487)	.530 (.827)
TWA	.050 (.045)	.102 (.067)	.104 (.066)	.106 (.085)
United	.034 (.044)	.044 (.041)	.046 (.042)	.053 (.065)
US Air	.249 (.108)	.210 (.087)	.211 (.090)	.378 (.193)
Western	.163 (.145)	.129 (.160)	.136 (.172)	.268 (.144)

* The numbers in parentheses are the standard deviations.

allocative inefficiency for each airline are not much different from zero.¹³ This can be seen from Table 4 where we report the mean (over time) values of costs of allocative inefficiency for each airline.

4. Conclusion

In this paper we focused on the specification of technical and allocative inefficiency using a translog cost function and the associated share equations. Using a formulation suggested by Schmidt (1984), the paper dealt with estimation of the cost function parameters and decomposition of cost-inefficiency into measures of technical and allocative inefficiencies. The main contribution of the paper is in providing a theoretically consistent linkage between allocative inefficiency—as measured by the errors of the cost share equations—and cost

¹³ As noted before the cost of allocative inefficiency, $\ln C_a$, as specified in (3) is an approximation. Since U_i are errors in the cost share equations, they are less than unity in absolute value. Thus $\ln C_a$ —which is measured by the weighted average of squared residuals of the share equations—may be biased towards zero.

therefrom. The methodology is applied to a sample of ten US airlines observed over 1970–8. The main results are: (i) none of the ten airlines was operating on the cost frontier in any of the years. Technical inefficiency increased (mean) cost by 4.19% for the best airline (Frontier) and 5.22% for the most inefficient airline (TWA). These estimates are quite robust to some alternative models. (ii) There is very small year-to-year variation in costs of technical inefficiency for these airlines. This finding is supported by all four models. (iii) Costs of allocative inefficiency are very low in all these models. Based on these results one may argue that these airlines were allocatively efficient during 1970–8.

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APPENDIX 1

The purpose of this appendix is to derive equation (7) in the text. This is done in the following two steps.

Step 1: Rewrite $q \equiv U'AU$. Since $\sum_{i=1}^n U_i = 0$, q can be expressed as $q = u'Ku$ where $u = (U_1, \dots, U_{n-1})'$ and K is a $(n-1) \times (n-1)$ symmetric matrix. Its elements k_{ij} are related to the elements of A , a_{ij} , in the following way

$$\begin{aligned} k_{ii} &= a_{ii} + a_{nn} - 2a_{in} \\ k_{ij} &= a_{ij} + a_{nn} - 2(a_{in} + a_{jn}), \quad i \neq j, \quad i = 1, \dots, n-1. \end{aligned}$$

Step 2: Differentiate q with respect to $\ln w_i$ to get

$$\frac{\partial q}{\partial \ln w_i} = \sum_j \frac{\partial q}{\partial U_j} \frac{\partial U_j}{\partial \ln w_i} = U_i. \quad (\text{A.1})$$

Now using

$$\frac{\partial q}{\partial U_j} = 2 \sum_{i=1}^{n-1} k_{ij} U_i$$

and

$$U_j = S_j - (\alpha_j + \sum_{i=1}^n \beta_{ij} \ln w_i + \sum_{k=1}^m d_{ik} \ln q_k + \lambda_j \ln y)$$

in (A.1) we get

$$\frac{\partial q}{\partial \ln w_i} = 2 \sum_j \left[\sum_l k_{lj} U_l (-\beta_{ij}) \right] = U_i. \quad (\text{A.2})$$

The relationship in (A.1) holds if

$$\sum_j \beta_{ij} k_{lj} = -\frac{1}{2} \quad (\text{when } l = i)$$

$$\sum_j \beta_{ij} k_{lj} = 0 \quad (\text{when } l \neq i).$$

APPENDIX 2
Parameter Estimates (Asymptotic Standard Errors in Parentheses)

Variable*	Model I	Model II	Model III	Model IV
Constant	14.244 (.012)	14.253 (.013)	14.368 (.011)	14.273 (.014)
Labor	.376 (.0024)	.375 (.003)	.375 (.003)	.379 (.004)
Fuel	.177 (.002)	.169 (.002)	.169 (.002)	.177 (.002)
Time	-.017 (.009)	-.016 (.008)	-.016 (.009)	-.012 (.002)
Stage Length	-.387 (.224)	-.371 (.104)	-.630 (.110)	-.318 (.147)
Load Factor	-.110 (.209)	-.365 (.199)	-.508 (.230)	-.223 (.166)
Points Served	-.027 (.106)	-.019 (.077)	-.124 (.082)	.043 (.081)
Output	.781 (.107)	.890 (.059)	1.00**	.769 (.077)
(Output) ²	.081 (.199)	.097 (.030)	—	.085 (.118)
(Labor)(Output)	.002 (.005)	—	—	-.002 (.005)
(Fuel)(Output)	.019 (.002)	—	—	.021 (.026)
(Time)(Output)	.001 (.010)	—	—	—
(Stage Length) (Output)	-.169 (.376)	—	—	-.212 (.219)
(Load Factor) (Output)	.421 (.293)	—	—	.413 (.207)
(Points Served) (Output)	.032 (.104)	—	—	-.076 (.161)
(Labor) ²	.136 (.017)	.136 (.017)	.131 (.018)	.106 (.020)
(Fuel) ²	.094 (.002)	.092 (.002)	.097 (.002)	.101 (.002)
(Labor)(Fuel)	-.029 (.002)	-.028 (.004)	-.028 (.004)	-.054 (.004)
(Time) ²	.000 (.003)	.001 (.003)	.004 (.003)	—
(Stage Length) ²	.221 (.727)	-.216 (.146)	.099 (.123)	.229 (.466)

Variable*	Model I	Model II	Model III	Model IV
(Load Factor) ²	.221 (2.757)	-3.507 (2.866)	2.310 (3.293)	-3.674 (.811)
(Points Served) ²	.161 (.303)	.054 (.206)	-.252 (.227)	.297 (.310)
(Points Served) (Stage Length)	.283 (.279)	.134 (.079)	.188 (.089)	.328 (.275)
(Point Served) (Time)	-.008 (.0014)	-.012 (.008)	-.012 (.009)	—
(Point Served) (Load Factor)	-.678 (.460)	-.401 (.274)	-.308 (.317)	-.988 (.256)
(Stage Length) (Time)	-.008 (.021)	-.019 (.008)	-.021 (.009)	—
(Stage Length) (Load Factor)	-.907 (.597)	-.129 (.256)	.109 (.286)	-.951 (.375)
(Load Factor) (Time)	-.025 (.087)	.006 (.083)	-.031 (.096)	—
(Labor) (Time)	-.006 (.001)	-.007 (.002)	-.007 (.002)	—
(Labor) (Stage Length)	-.043 (.010)	-.039 (.003)	-.039 (.003)	-.030 (.010)
(Labor) (Load Factor)	.061 (.034)	.060 (.034)	.060 (.034)	-.028 (.029)
(Labor) (Points Served)	.027 (.008)	.029 (.006)	.029 (.006)	.034 (.008)
(Fuel)(Time)	.001 (.001)	.002 (.001)	.002 (.001)	—
(Fuel) (Stage Length)	-.008 (.005)	.028 (.002)	.028 (.002)	-.013 (.005)
(Fuel) (Load Factor)	-.070 (.017)	-.090 (.021)	-.087 (.022)	-.049 (.015)
(Fuel) (Points Served)	-.029 (.004)	-.008 (.004)	-.008 (.004)	-.031 (.004)
σ_v	.315 (.013)	.299 (.014)	.234 (.013)	.272 (.011)
σ_τ	.057 (.024)	.063 (.023)	.065 (.030)	.074 (.036)
σ_{11}	$.209 \times 10^{-3}$ ($.477 \times 10^{-4}$)	$.212 \times 10^{-3}$ ($.492 \times 10^{-4}$)	$.212 \times 10^{-3}$ ($.496 \times 10^{-4}$)	$.237 \times 10^{-3}$ ($.599 \times 10^{-4}$)
σ_{12}	$.248 \times 10^{-5}$ ($.446 \times 10^{-5}$)	$.517 \times 10^{-5}$ ($.492 \times 10^{-5}$)	$.596 \times 10^{-5}$ ($.496 \times 10^{-5}$)	$.194 \times 10^{-5}$ ($.435 \times 10^{-5}$)
σ_{22}	$.519 \times 10^{-4}$ ($.795 \times 10^{-5}$)	$.837 \times 10^{-4}$ ($.151 \times 10^{-4}$)	$.853 \times 10^{-4}$ ($.155 \times 10^{-4}$)	$.583 \times 10^{-4}$ ($.936 \times 10^{-5}$)

* All the variables, except time, are in logs. Labor and Fuel refer to their prices.

** Constrained to unity

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